CSCE 5300 – Introduction to Big Data and Data Science

Assignment 2: Coding Structured Data Processing

1. Problem Statement

The last digit of my UNT Canvas student login ID is 0. Per the assignment instructions, this value maps to the SELECT statement category (n = 0, 3, 6, or 9). The objective of this assignment is to design an original relational data set, unrelated to any textbook example, and to write a series of SQL SELECT statements that demonstrate increasing levels of query complexity: basic column retrieval, conditional filtering combined with sorting, and a second conditional filter based on a text value rather than a numeric one.

The data set represents appointment records for a small mobile hair styling business. Each record captures a single client booking, including the type of service performed, its price, the date of the appointment, and the location where the service took place. This scenario was chosen because it reflects a realistic, small-scale business operation where SQL queries would genuinely be used to track revenue and client activity.

2. Method

The approach followed the structure introduced in Chapter 2 of the CSCE 5300 course material, which presents the SELECT statement as the foundational mechanism for retrieving data from a relational table. Three original query examples were developed to mirror the progression covered in the chapter:

1. A basic SELECT statement that retrieves a subset of columns without any filtering.

2. A SELECT statement combined with a WHERE-style condition and an ORDER BY clause, isolating higher-priced services and sorting the results.

3. A second SELECT statement demonstrating a string-based WHERE condition, isolating all appointments of a single service type rather than a numeric threshold.

A table named SalonBookings was created with six columns: booking_id, client_name, service_type, service_price, booking_date, and venue_location. Ten original rows of sample data were manually authored to populate the table before running the queries. The code was executed and validated in a SQLite environment to confirm correctness prior to inclusion in this report.

3. Code

Table creation and sample data:

```
CREATE TABLE SalonBookings (
    booking_id      INT PRIMARY KEY,
    client_name     VARCHAR(50),
    service_type    VARCHAR(30),
    service_price   DECIMAL(6,2),
    booking_date    DATE,
    venue_location  VARCHAR(50)
);
```

```
INSERT INTO SalonBookings (booking_id, client_name, service_type,
    service_price, booking_date, venue_location) VALUES
(1, 'Dana Ruiz',      'Haircut',   45.00, '2026-08-01', 'Riverside Heights'),
(2, 'Marcus Lee',     'Color',     95.00, '2026-08-02', 'Riverside Heights'),
(3, 'Priya Anand',    'Braids',    120.00,'2026-08-03', 'Client Home Visit'),
(4, 'Sam Whitfield',  'Haircut',   45.00, '2026-08-05', 'Riverside Heights'),
(5, 'Olivia Chen',    'Blowout',   35.00, '2026-08-06', 'Client Home Visit'),
(6, 'Trevor Banks',   'Color',     100.00,'2026-08-07', 'Riverside Heights'),
(7, 'Nadia Farouk',   'Braids',    130.00,'2026-08-08', 'Client Home Visit'),
(8, 'Casey Odom',     'Haircut',   40.00, '2026-08-09', 'Riverside Heights'),
(9, 'Ellis Brandt',   'Blowout',   35.00, '2026-08-10', 'Client Home Visit'),
(10,'Renee Castillo', 'Color',     105.00,'2026-08-11', 'Riverside Heights');
```

Example 1 – Basic SELECT statement:

```
SELECT client_name, service_type
FROM SalonBookings;
```

Example 2 – SELECT with filtering and sorting:

```
SELECT client_name, service_type, service_price
FROM SalonBookings
WHERE service_price > 50.00
ORDER BY service_price DESC;
```

Example 3 – SELECT with a string-based filter:

```
SELECT client_name, service_type, service_price
FROM SalonBookings
WHERE service_type = 'Braids';
```

4. Results

The code was executed successfully against the sample data set. The output of each query is shown below.

Example 1 output (client and service type for all ten bookings):

client_name	service_type
Dana Ruiz	Haircut
Marcus Lee	Color
Priya Anand	Braids
Sam Whitfield	Haircut
Olivia Chen	Blowout
Trevor Banks	Color
Nadia Farouk	Braids
Casey Odom	Haircut
Ellis Brandt	Blowout
Renee Castillo	Color

Example 2 output (services priced above $50, sorted highest to lowest):

client_name	service_type	service_price
Nadia Farouk	Braids	130.00
Priya Anand	Braids	120.00
Renee Castillo	Color	105.00
Trevor Banks	Color	100.00
Marcus Lee	Color	95.00

Example 3 output (all Braids appointments):

client_name	service_type	service_price
Priya Anand	Braids	120.00
Nadia Farouk	Braids	130.00

The results confirm that the queries behaved as expected. The basic SELECT returned all ten client-service pairs without alteration. The numeric filter correctly excluded the four services priced at or below $50 and ordered the remaining six results from highest to lowest price. The string-based filter correctly isolated the two Braids appointments from the full data set, demonstrating that the WHERE condition works the same way for text values as it does for numeric ones, provided the value is enclosed in single quotes.

5. Conclusion

This assignment demonstrated the practical application of the SQL SELECT statement across three variations: direct column retrieval, numeric filtering with sorting, and string-based filtering. Using an original business scenario reinforced how relational queries translate directly into actionable business insight, such as identifying which clients booked a specific service at a small mobile styling operation.